INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

Module 4: Forecasting and Cancellation Trends

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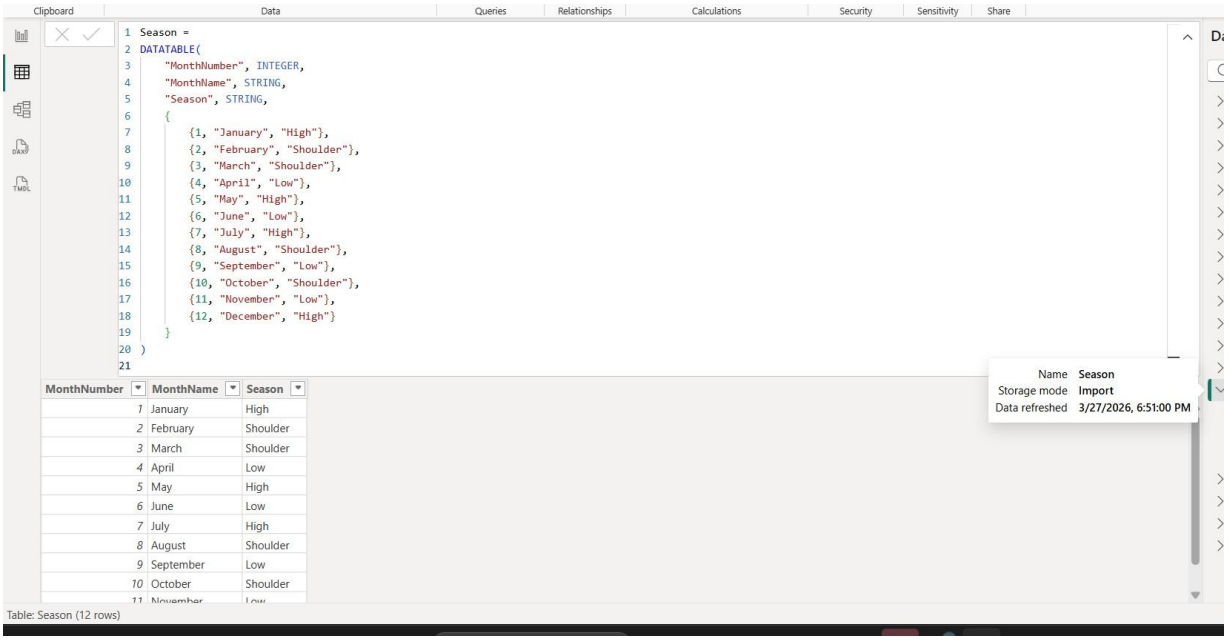
PROJECT NAME: HotelRevAI AI Driven Revenue Analysis for Hotels_Batch -13

Creating season table

1. Open Power BI → Modeling tab → Click "New Table"
2. Paste the given DATATABLE code in the formula bar

```
Season =
DATATABLE(
  "MonthNumber", INTEGER,
  "MonthName", STRING,
  "Season", STRING,
  {
    {1, "January", "High"},
    {2, "February", "Shoulder"},
    {3, "March", "Shoulder"},
    {4, "April", "Low"},
    {5, "May", "High"},
    {6, "June", "Low"},
    {7, "July", "High"},
    {8, "August", "Shoulder"},
    {9, "September", "Low"},
    {10, "October", "Shoulder"},
    {11, "November", "Low"},
    {12, "December", "High"}
  }
)
```

3. Press Enter to create the table
4. Go to Data View to verify the Season table is created

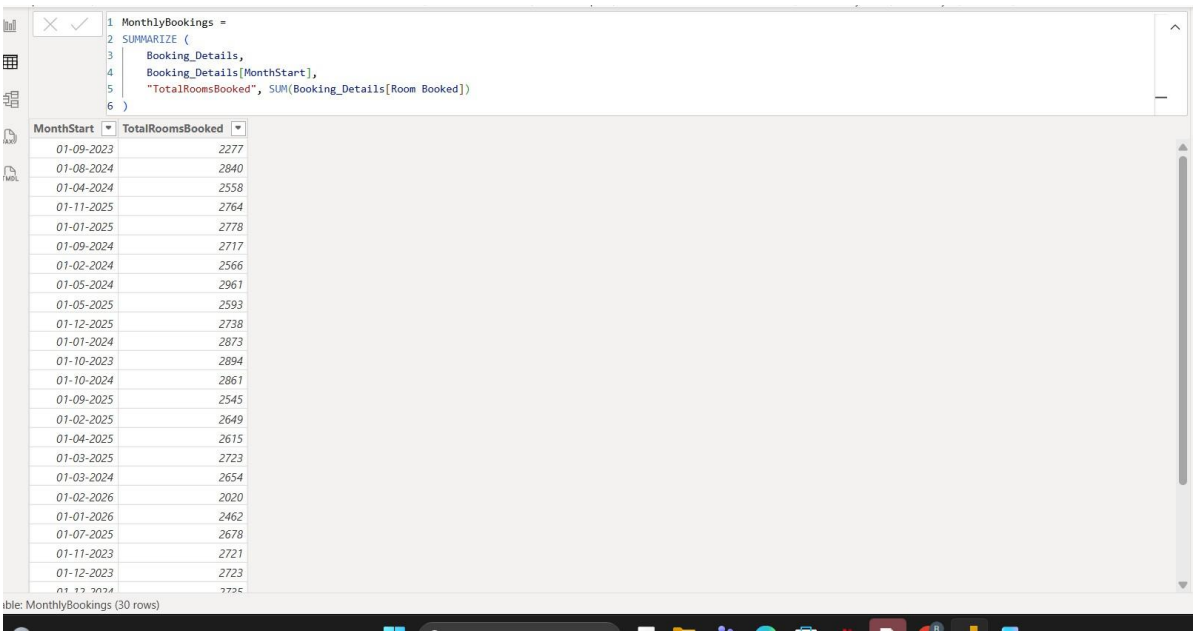


Creating Monthly bookings

1. Open Power BI → Modeling tab → Click "New Table"
2. Paste the given DATATABLE code in the formula bar

```
MonthlyBookings =
SUMMARIZE (
   Booking_Details,
   Booking_Details[MonthStart],
   "TotalRoomsBooked", SUM(Booking_Details[Room Booked])
)
```

3. Press Enter to create the table
4. Go to Data View to verify the Monthly bookings table is created



Add New Columns in Bookings Table

1. Go to **Data View** in Power BI
2. Select the **Bookings table**
3. Click on **New Column**
4. Paste the formula → Press **Enter**
5. Repeat the same steps for each column below

Columns to Create

CancellationFlag

```
CancellationFlag =
IF(Bookings[BookingStatus] = "Cancelled", "Yes", "No")
```

NoShowFlag

```
NoShowFlag =
IF(Bookings[BookingStatus] = "No-Show", 1, 0)
```

RefundAmount

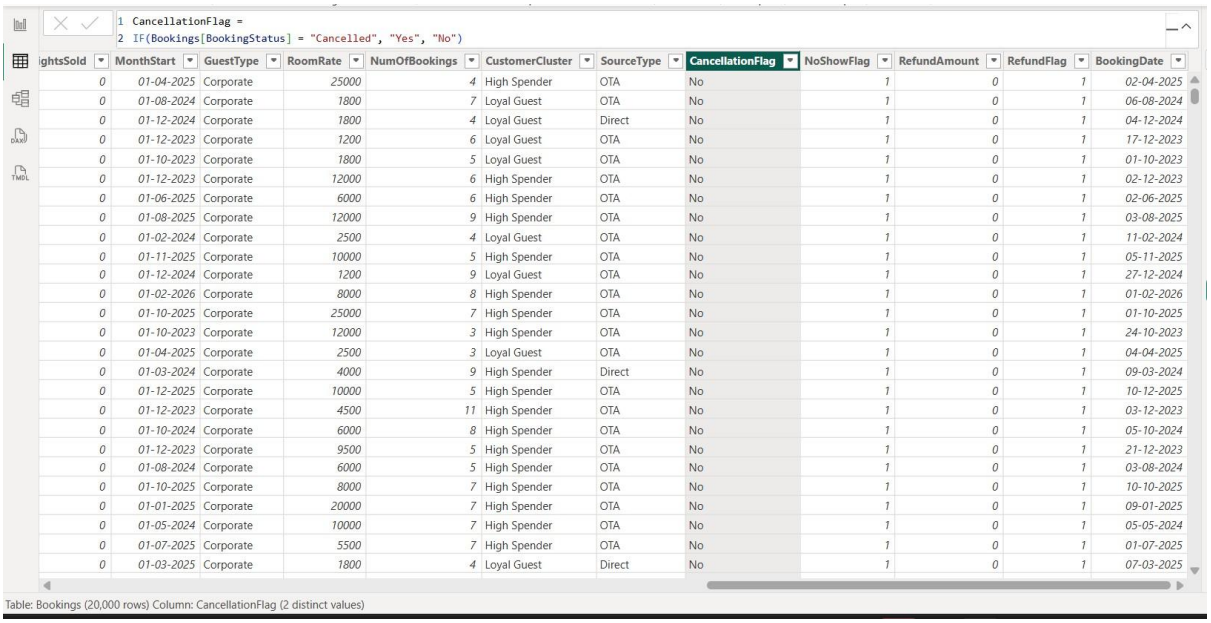
```
RefundAmount =
IF(
  Bookings[BookingStatus] = "Cancelled",
  Bookings[Revenue],
  0
)
```

RefundFlag

```
RefundFlag =
IF(Bookings[RefundAmount] > 0, 1, 0)
```

BookingDate

```
BookingDate =
VAR CheckInDate = Bookings[CheckInDate]
VAR PriorMonthDate = EOMONTH(CheckInDate, -1) + 1
VAR DaysBetween = DATEDIFF(PriorMonthDate, CheckInDate, DAY)
VAR RandomOffset = RANDBETWEEN(0, DaysBetween)
RETURN PriorMonthDate + RandomOffset
```



```
CancellationFlag =
IF(Bookings[BookingStatus] = "Cancelled", "Yes", "No")
```

ightsSold	MonthStart	GuestType	RoomRate	NumOfBookings	CustomerCluster	SourceType	CancellationFlag	NoShowFlag	RefundAmount	RefundFlag	BookingDate
0	01-04-2025	Corporate	25000	4	High Spender	OTA	No	1	0	1	02-04-2025
0	01-08-2024	Corporate	1800	7	Loyal Guest	OTA	No	1	0	1	06-08-2024
0	01-12-2024	Corporate	1800	4	Loyal Guest	Direct	No	1	0	1	04-12-2024
0	01-12-2023	Corporate	1200	6	Loyal Guest	OTA	No	1	0	1	17-12-2023
0	01-10-2023	Corporate	1800	5	Loyal Guest	OTA	No	1	0	1	01-10-2023
0	01-12-2023	Corporate	12000	6	High Spender	OTA	No	1	0	1	02-12-2023
0	01-06-2025	Corporate	6000	6	High Spender	OTA	No	1	0	1	02-06-2025
0	01-08-2025	Corporate	12000	9	High Spender	OTA	No	1	0	1	03-08-2025
0	01-02-2024	Corporate	2500	4	Loyal Guest	OTA	No	1	0	1	11-02-2024
0	01-11-2025	Corporate	10000	5	High Spender	OTA	No	1	0	1	05-11-2025
0	01-12-2024	Corporate	1200	9	Loyal Guest	OTA	No	1	0	1	27-12-2024
0	01-02-2026	Corporate	8000	8	High Spender	OTA	No	1	0	1	01-02-2026
0	01-10-2025	Corporate	25000	7	High Spender	OTA	No	1	0	1	01-10-2025
0	01-10-2023	Corporate	12000	3	High Spender	OTA	No	1	0	1	24-10-2023
0	01-04-2025	Corporate	2500	3	Loyal Guest	OTA	No	1	0	1	04-04-2025
0	01-03-2024	Corporate	4000	9	High Spender	Direct	No	1	0	1	09-03-2024
0	01-12-2025	Corporate	10000	5	High Spender	OTA	No	1	0	1	10-12-2025
0	01-12-2023	Corporate	4500	11	High Spender	OTA	No	1	0	1	03-12-2023
0	01-10-2024	Corporate	6000	8	High Spender	OTA	No	1	0	1	05-10-2024
0	01-12-2023	Corporate	9500	5	High Spender	OTA	No	1	0	1	21-12-2023
0	01-08-2024	Corporate	6000	5	High Spender	OTA	No	1	0	1	03-08-2024
0	01-10-2025	Corporate	8000	7	High Spender	OTA	No	1	0	1	10-10-2025
0	01-01-2025	Corporate	20000	7	High Spender	OTA	No	1	0	1	09-01-2025
0	01-05-2024	Corporate	10000	7	High Spender	OTA	No	1	0	1	05-05-2024
0	01-07-2025	Corporate	5500	7	High Spender	OTA	No	1	0	1	01-07-2025
0	01-03-2025	Corporate	1800	4	Loyal Guest	Direct	No	1	0	1	07-03-2025

Table: Bookings (20,000 rows) Column: CancellationFlag (2 distinct values)

Steps to Add Measures in Bookings Table

1. Go to **Report View or Data View**

2. In the Fields pane, select **Bookings table**
3. Click on **New Measure**
4. Paste one formula and press **Enter**
5. Repeat the same steps for each measure

Measures to Create

1. Bookings

```
Bookings =
COUNTROWS(Bookings)
```

2. CancellationRate

```
CancellationRate =
DIVIDE(
  COUNTROWS(
    FILTER(Bookings, Bookings[CancellationFlag] = "Yes")
  ),
  COUNTROWS(Bookings)
)
```

3. NoShowCount

```
NoShowCount =
CALCULATE(
  COUNTROWS(Bookings),
  Bookings[BookingStatus] = "No-Show"
)
```

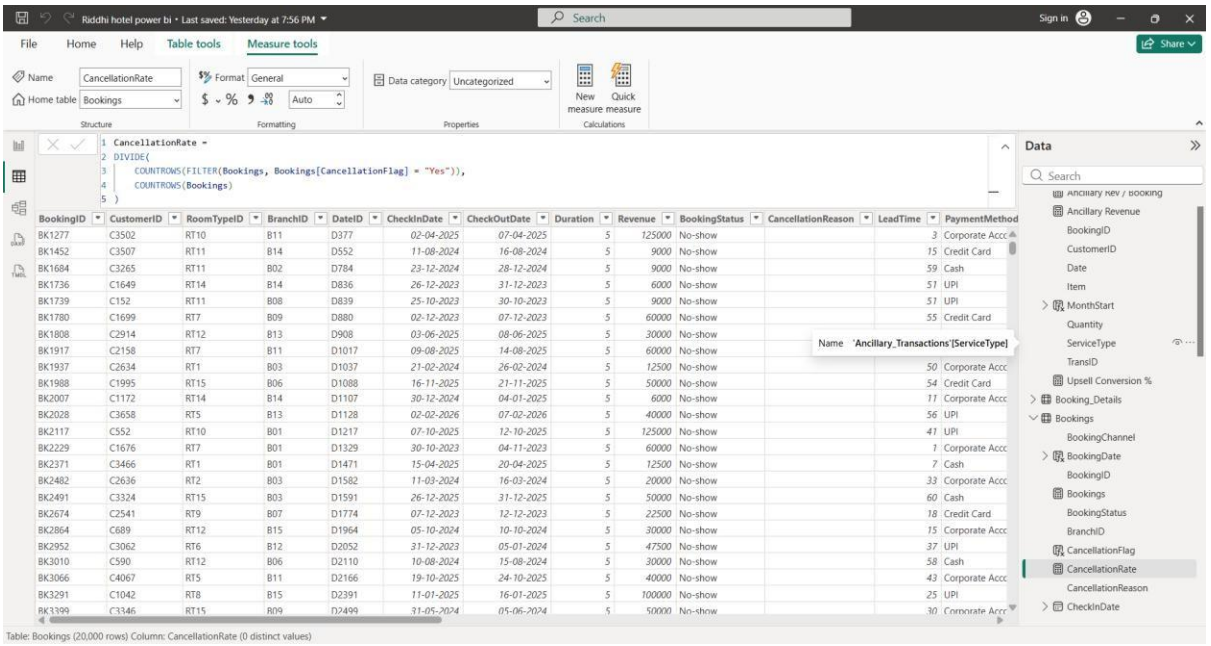
4. Recommended Price

```
Recommended Price =
VAR BaseADR = [DADR]
VAR SeasonAdj =
  SWITCH(
    SELECTEDVALUE('Date'[Season]),
    "High", 0.15,
    "Shoulder", 0.05,
    "Low", -0.10,
    0
  )
VAR Uplift = [Price Uplift % Value]
VAR Elasticity = [Occupancy Sensitivity Value]
```

```
RETURN
   BaseADR * (1 + SeasonAdj + Uplift) * Elasticity
```

5. RefundCount

```
RefundCount =
CALCULATE(
   COUNTROWS(Bookings),
   Bookings[RefundAmount] > 0
)
```



Creating Visuals

Steps to Create Line Chart with Forecast

Step 1: Insert Line Chart

1. Select Line Chart
2. Drag:
   - StayDate → X-axis
   - Occupancy % → Y-axis
3. Click dropdown on Occupancy → set to Average

Step 2: Fix X-Axis (Very Important)

1. Click on StayDate (X-axis)
2. Select:
   - StayDate (not Date hierarchy)

3. Go to Format → X-axis
    - Type → Continuous

Step 3: Basic Formatting

1. Go to Format
2. Turn ON:
    - Markers
    - Data labels (optional)
3. Add title:
    - *Average Occupancy by StayDate*

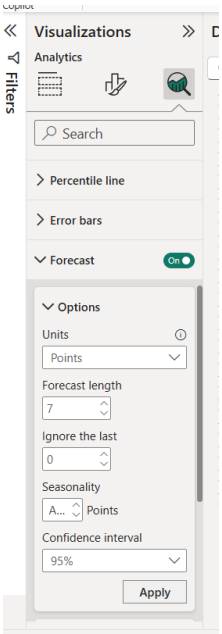
Step 4: Add Forecast

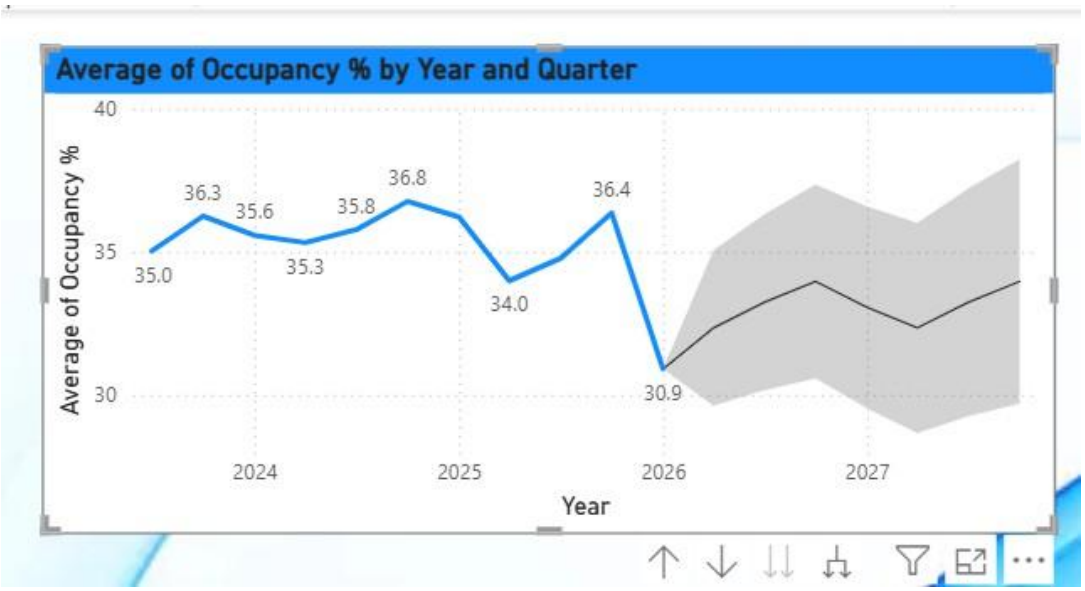
1. Select the line chart
2. Open Analytics pane (magnifying glass icon)
3. Find Forecast → Click Add

Step 5: Configure Forecast

Set:

- Forecast length → 3 or 6 Months
- Confidence interval → 95%
- Seasonality → Auto (or 12 for monthly data)
- Ignore last → 0





2. Cancellation Rate by MonthStart (Line Chart)

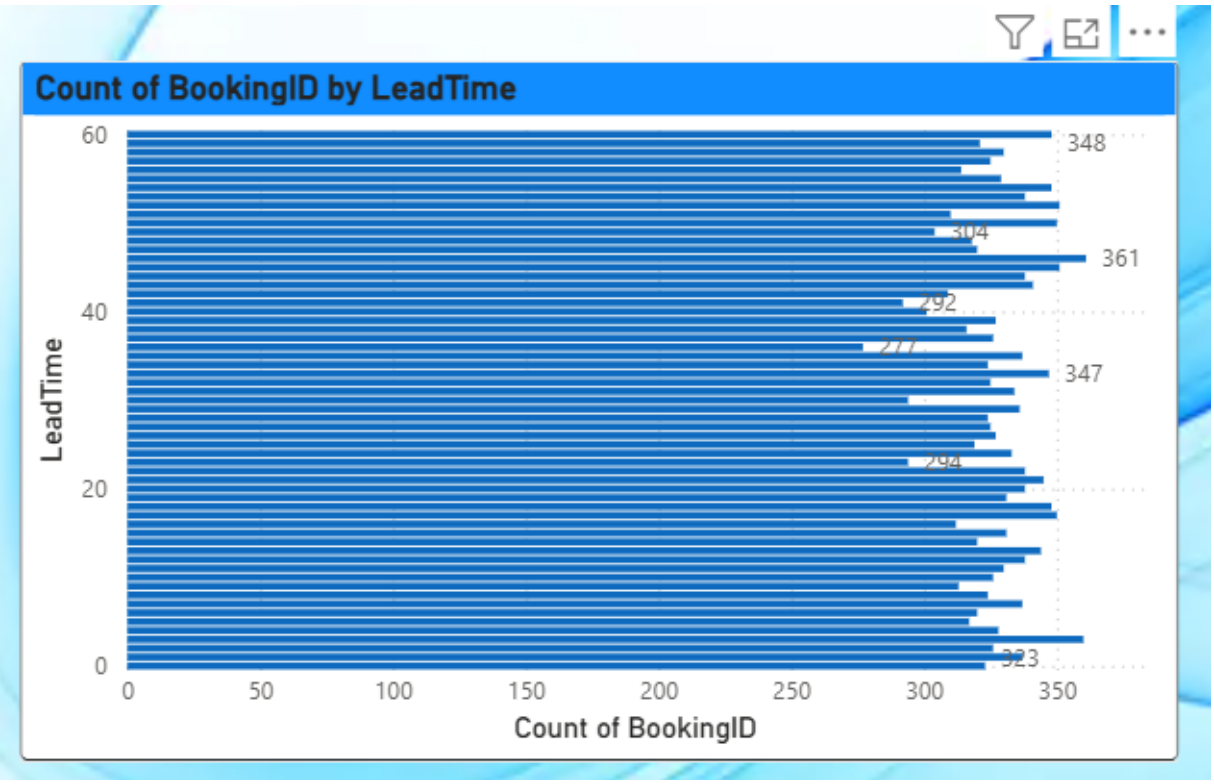
Steps:

1. Select **Line Chart**
2. Drag:
   - **Axis** → MonthStart
   - **Values** → CancellationRate (measure)
3. Format:
   - Turn ON markers
   - Title: *CancellationRate by MonthStart*

3. Count of BookingID by LeadTime (Bar Chart)

Steps:

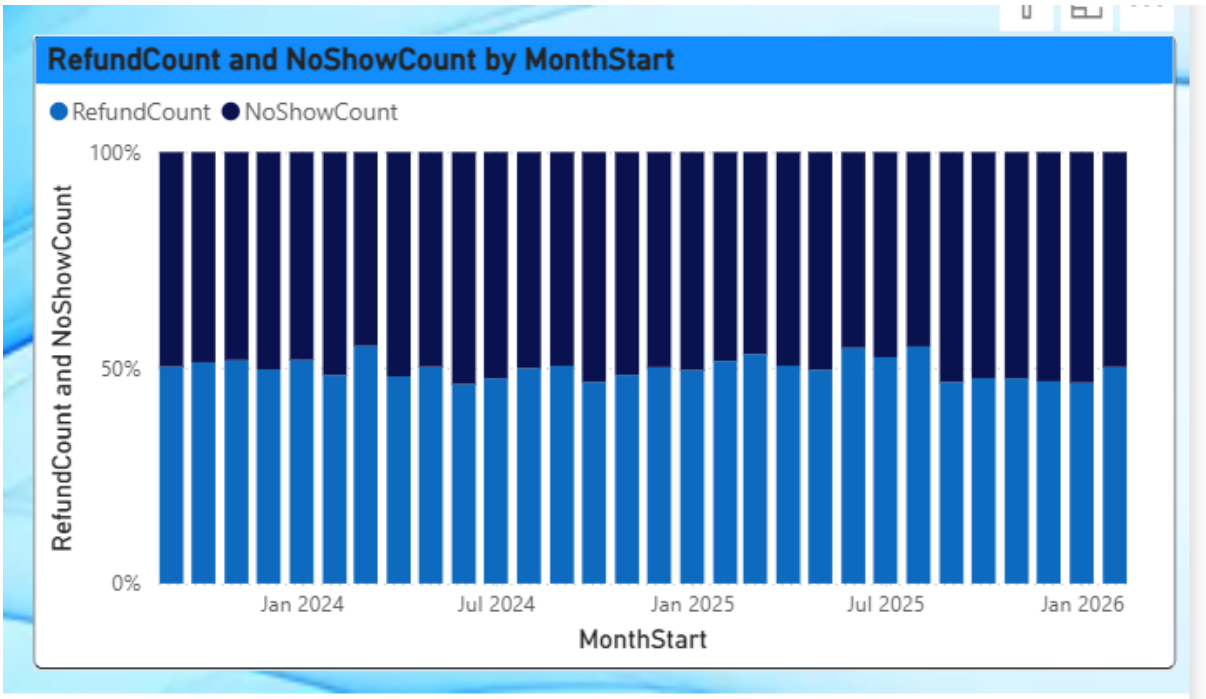
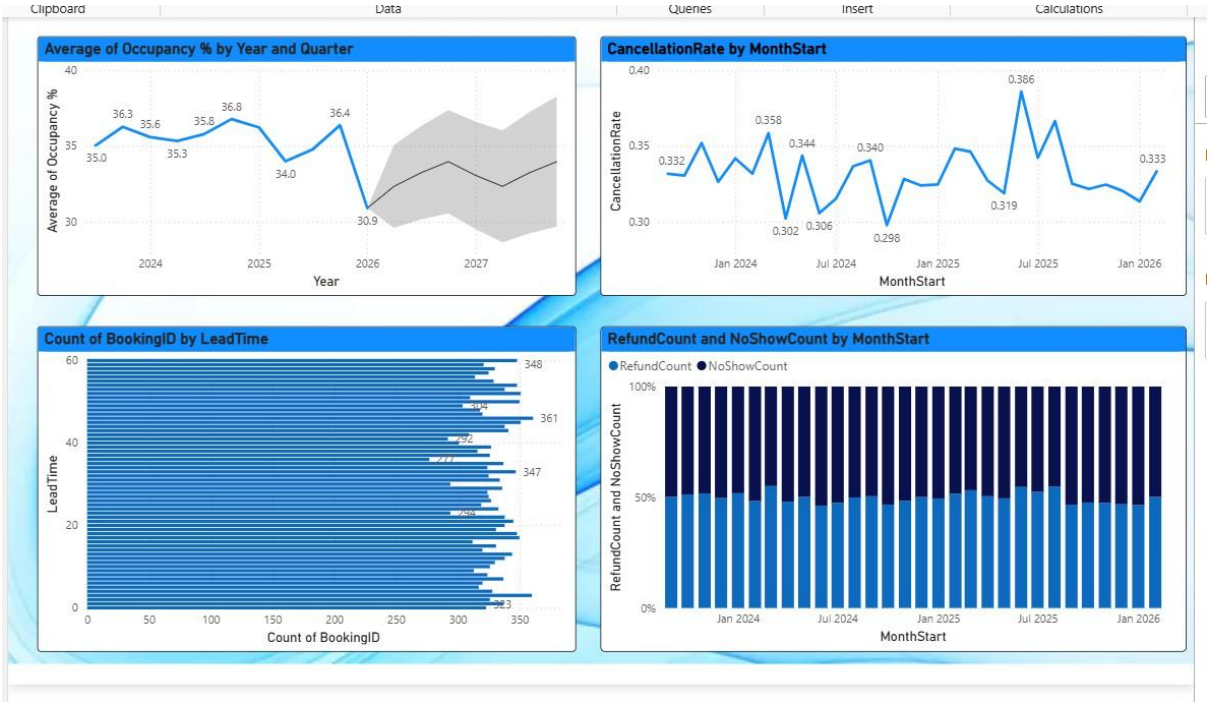
1. Select **Clustered Bar Chart**
2. Drag:
   - **Y-axis** → LeadTime
   - **X-axis** → BookingID (Count)
3. Sort:
   - Click on visual → Sort by BookingID (Descending)
4. Title: *Count of BookingID by LeadTime*



4. RefundCount and NoShowCount by MonthStart (Stacked Column Chart)

Steps:

1. Select **Stacked Column Chart**
2. Drag:
   - **Axis** → MonthStart
   - **Values** → RefundCount, NoShowCount
3. Format:
   - Turn ON **Legend**
   - Title: *RefundCount and NoShowCount by MonthStart*



Final Dashboard